

Ali Awil

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EDUCATION

University of Ottawa <i>Master of Computer Science, Applied Artificial Intelligence</i>	Jan 2026 – Jun 2027 Ottawa
• Coursework: Machine Learning, Virtual Assistants, Introduction to Deep Learning and Reinforcement Learning, Ethics in AI	
University of Ottawa <i>Bachelor of Computer Science, Data Science Option (CO-OP)</i>	Jan 2021 – Dec 2025 Ottawa

SKILLS

- **Languages:** Python, Java, C++, JavaScript, TypeScript, SQL, R
- **Frameworks / Tools:** React, LangChain, Flask, FastAPI, PyTorch, Scikit-learn, Hugging Face, Power BI, Docker, Azure, Git
- **Concepts:** Machine Learning, Model Evaluation, RAG, Agentic AI, Vector Databases, Cloud Deployment, Data Visualization

EXPERIENCE

University of Ottawa <i>Teaching Assistant</i>	Jan 2026 – Present Ottawa, Canada
• Teaching Assistant for CSI 3131 (Operating Systems), supporting students through labs and assignments in C and C++. • Assisted students with core OS concepts including processes, threads, synchronization, and memory management. • Guided students through debugging in systems programming and reinforced understanding during office hours and lab sessions.	
Department of National Defence <i>Data Science Developer Intern</i>	May 2024 – May 2025 Ottawa, Canada
• Optimized model selection for FAM policy Q&A; by benchmarking accuracy and latency across Longwizard-70B and Mistral Nemo, resulting in 23% faster inference and 89% accuracy. • Tested a fine-tuned LLaVA (WizardLM-70B + CLIP) model and analyzed outputs to evaluate multimodal reasoning quality. • Contributed to two large-scale AI research initiatives (LLaVA multimodal and Space News), leading end-to-end experimentation, benchmarking, and documentation. • Evaluated the Space News information extraction pipeline, validating entity-level accuracy and improving extraction reliability.	
Zafin <i>Software Engineer Intern</i>	May 2023 – Aug 2023 Toronto, Canada
• Developed an Enterprise RAG Knowledge Assistant using OpenAI's API that achieved 90% accuracy on internal benchmarks. • Architected a custom RAG pipeline by ingesting and processing 10K+ internal documents into a PostgreSQL vector database. • Evaluated performance/cost tradeoffs between open-source (Falcon-7B) and proprietary (GPT) models; deployed the solution via a Flask microservice on Azure. • Co-authored "Closing the Gap Between Available Information and Accessible Knowledge", a company research article published on Zafin's Engineering Blog (Medium, Aug 2023).	

PROJECTS

Voice Assistant (CSI5180 – Topics in AI)

- Built an end-to-end VA pipeline: speaker verification (ECAPA-TDNN), wake word detection, Whisper ASR, and fine-tuned BERT for joint intent classification and slot filling.
- Integrated OMDb and MusicBrainz APIs for real-time fulfillment across movie and music domains with a Smart Media Center control system handling 10 playback intents.
- Applied BIO tagging for slot extraction across 20+ intents spanning movies, music, and system control domains.

Flight Price Prediction (CSI5155 – Machine Learning)

- Trained and compared Random Forest, XGBoost, and a TensorFlow DNN on 300K+ flight records; DNN achieved $R^2 = 0.97$ on the full dataset, outperforming tree-based baselines.
- Applied SHAP for model explainability and validated statistical significance of DNN improvements via paired t-tests ($p \approx 0$).